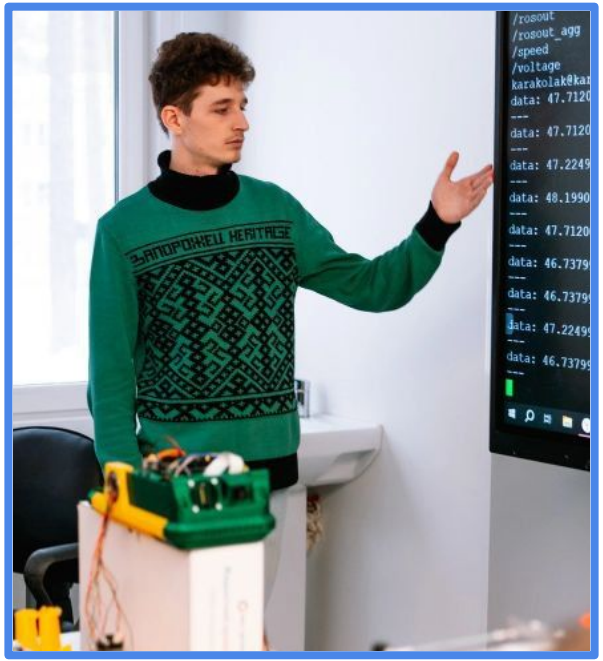
What to do if you take part in ORBITA Tournament?

Tips from the organizers

The Experts



Boris Nikitin



Nikita Vanag

Correspondence part of the tournament

1. Onboarding (👋 Start Here)

- Meet the organizers online
- Read the rules and ask questions



3. Build Your Project🔧

- Write code / assemble hardware
- Test and improve



2. Pick Your Experiment🧪

- Come up with your own experiment idea
- Discuss its feasibility with experts



About stratospheric launch

Altitude and Flight Conditions

- **Operating altitude:** 20-35 km (Karman line - space boundary - is 100 km)
- **Atmospheric pressure:** <1% of sea level (similar to Low Earth Orbit conditions)
- **Temperature:** -70°C to +20°C
- **Radiation:** 100-200× stronger than at sea level



Stratospheric Flight Testing

(Flight Stages)



1. **Ascent (60-90 min):**
   - Speed: 5 m/s (~18 km/h)
   - Gradual atmospheric thinning → tests for pressure integrity and thermal regulation
2. **Peak altitude (10-15 min):**
   - Balloon bursts
   - Equipment operates in near-space conditions
3. **Descent (30-40 min):**
   - Parachute deployment G-forces
   - Vibration and turbulence
   - Possible water/swamp landing

Recommendations for assigning roles in a team

- **Project Manager** - Oversees the entire project, coordinates the work of the team and controls deadlines.
- **Mechanical Engineer** - Designs and integrates the payload and subsystems into the CubeSat frame.
- **Electrical Engineer** - Develops the power system and soldering onboard electronics.
- **Embedded Software Engineer** - Develops and tests flight software for data collection, communication, and control.
- **Radio Communication Engineer.** Develops software for the spacecraft radio module and ground station, ensuring reliable and uninterrupted communication with the CubeSat satellite during flight.
- **Theoretical Analysis & Presentation Specialist** - Responsible for the scientific foundation of the experiment including theoretical modeling, data interpretation and presentation.

General recommendations for choosing an experiment



1. Follow Your Passion/Expertise

Pick a field you enjoy (e.g., biology, physics, engineering).

Example: Study plant/microorganism adaptation to near-space conditions.



2. Research Existing Work

Avoid duplication: Check if similar experiments exist.

Improve prior designs: Add new variables or measurement methods.

EASY

3. Prioritize Simplicity & Feasibility

Better a completed simple experiment than an unfinished complex one.

Account for time constraints.



4. Leverage Unique Stratospheric Conditions

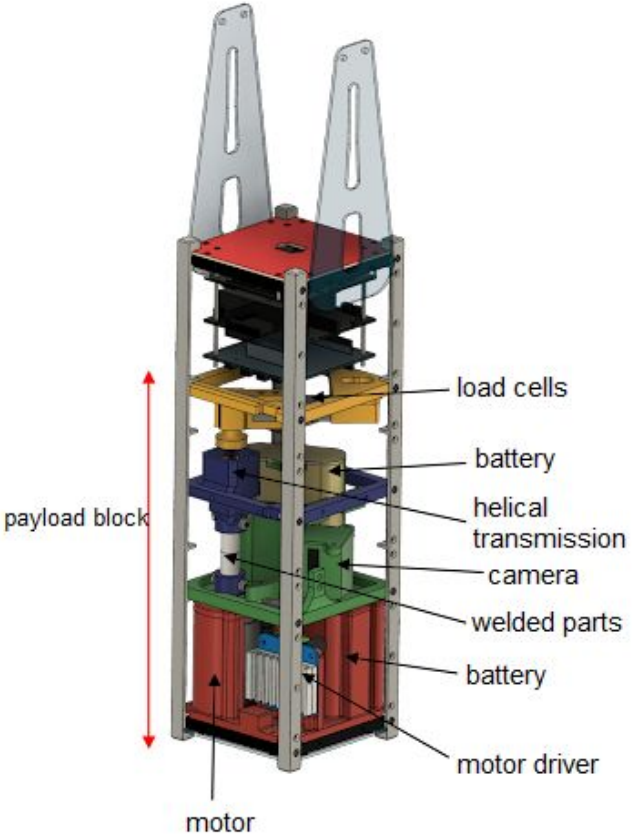
Focus on phenomena that require near-space environment. Avoid trivial tests (e.g., water freezing) – use a freezer instead!



5. Consider Scalability

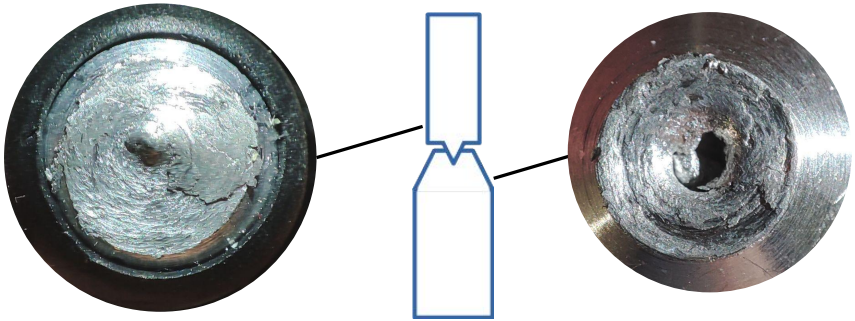
Tests with spaceflight potential get extra attention.

Examples of experiments

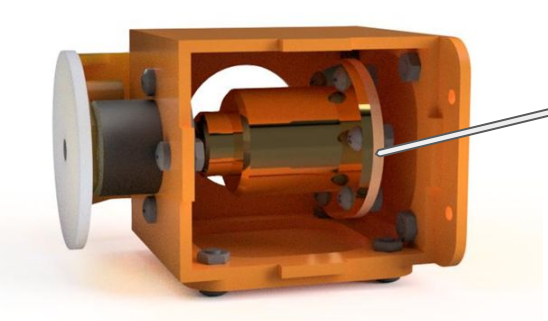




Friction welding aboard the CubeSat 3U prototype



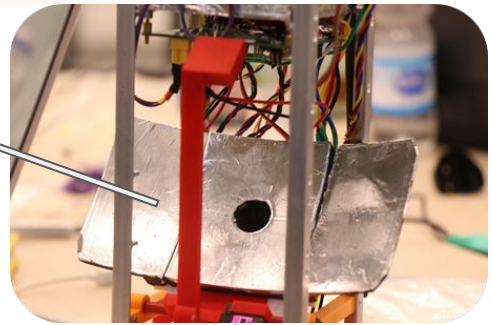
Development of metal alloy production technology using concentrated solar radiation on a 3U Cubesat platform

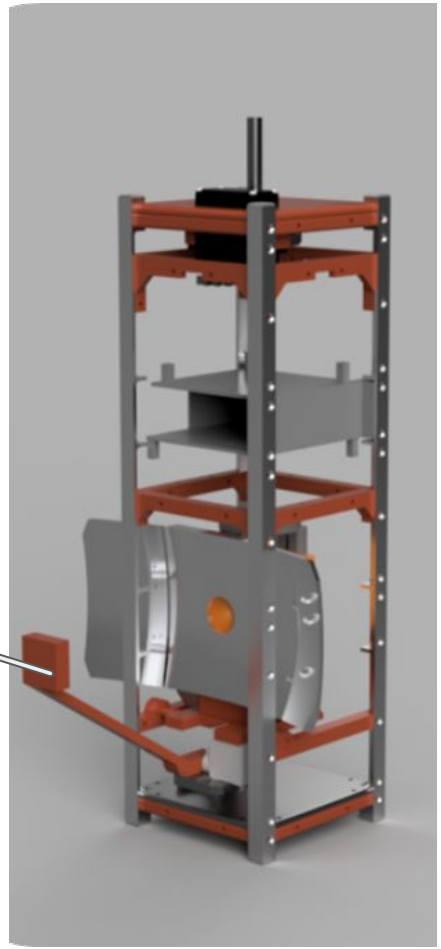


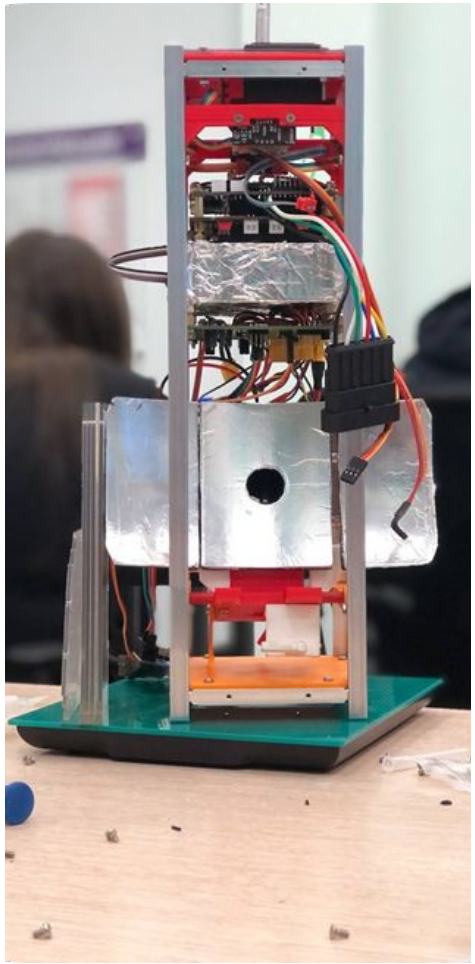
rotating heating chamber

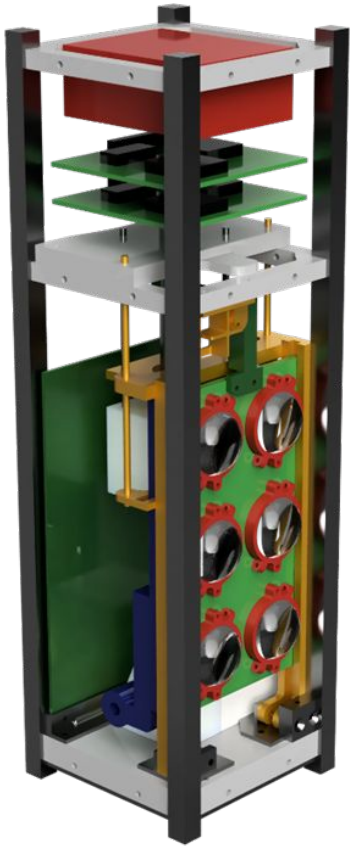
secondary reflector

main reflector



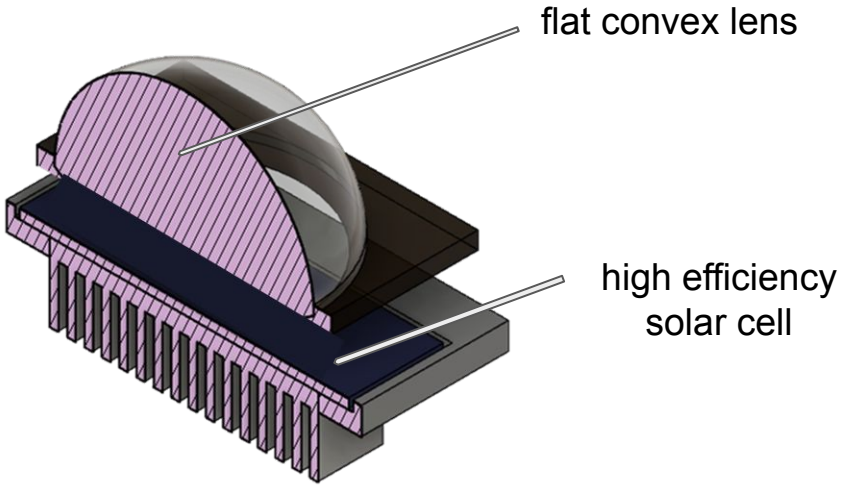




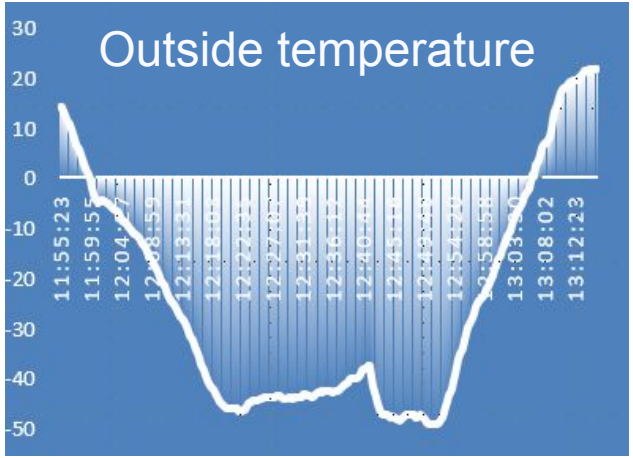




Design and development of a 3U Cubesat prototype equipped with gimbaled solar concentrator panels



And many others…

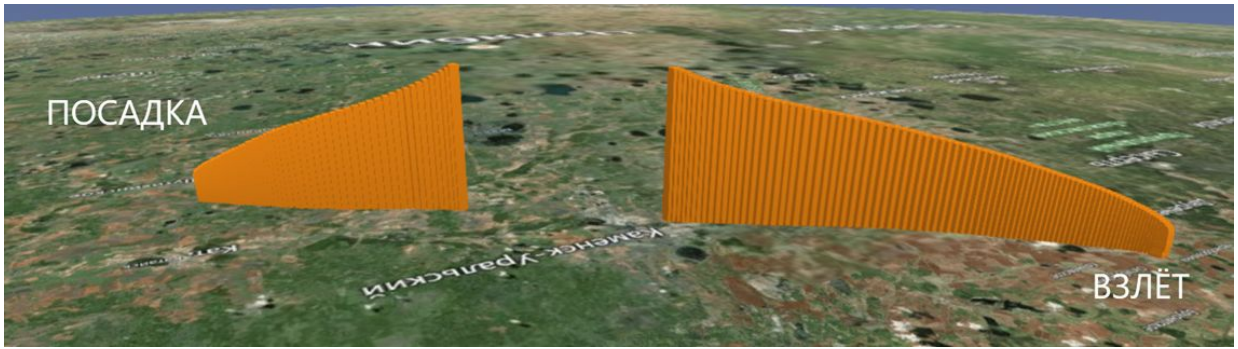




Picture from Cubesat



Google Maps



Evaluation criteria for final projects

Participant Guide for the Orbita Tournament

Критерии оценки финальных проектов

Руководство для участников турнира Орбита